

MAPQUEST: Hilbert-Fair Navigation

From Addressable Inferences to a Verified Target

Positive Completeness, Target Cones, and Near-Target Guidance

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Working research draft

September 2026

Abstract

This paper isolates a rigorous core of the MAPQUEST automated-theorem-proving program. Its central claim is conditional positive completeness: if a fixed target T has a finite formal proof in an effective finitary proof system, then a MAPQUEST execution that assigns finite addresses to all finite inference candidates, revisits those candidates fairly as the verified BANK grows, and admits conclusions only through a total sound verifier must reach a verified proof of T after finitely many search events. An explicit surjection from natural numbers onto rational addresses, together with tuple pairing, provides a concrete road atlas for formulas, ordered premise tuples, rule tags, and finite instantiation witnesses. Discrete Hilbert traversals then organize finite rule-specific address boxes without changing their coverage. The crucial additional ingredient is fair recurrence across both address resolution and BANK epochs; a single Hilbert pass is not enough because an inference can be encountered before its premises have been proved. The paper further develops backward target cones, waiting inference frames, a residual-obligation measure for nearness to T , and a two-queue architecture in which Hilbert-fair enumeration provides the completeness backbone while target-conditioned MAPQUEST guidance, the Professor, Dreamer, oracle interfaces, Tenneson1, and Depth Charge attempt to reduce discovery time. Several diagrams make explicit the distinction between the complete search backbone and the target-directed acceleration layer.

Central theorem-shaped statement.

$$\begin{aligned} &\text{finite explicit inference addresses} + \text{effective decoding} + \text{total sound verification} \\ &+ \text{fair revisitation after BANK growth} \implies \Gamma \vdash T \Rightarrow \text{MAPQUEST eventually verifies } T. \end{aligned}$$

The Hilbert curve contributes an organized locality-preserving traversal of finite address regions; fairness, not geometry alone, supplies the completeness guarantee.

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1 The problem is to prove the fixed target T

The purpose of MAPQUEST is not to explore an abstract address space for its own sake. A run begins with a fixed formal environment, an initial verified set Γ , and a fixed target T . The objective is

$$\boxed{\Gamma \vdash T.}$$

All geometry, rational addressing, Hilbert scheduling, target heuristics, Professor scores, Dreamer proposals, oracle interventions, Depth Charges, and resource controls are subordinate to the construction of a verifier-accepted proof whose final conclusion is exactly T .

Let

$$\mathfrak{F} = (\Sigma, \mathcal{W}, \Gamma, \mathcal{R}, \mathcal{V})$$

be a frozen proof environment. The verified BANK at time t is $B_t \subseteq \mathcal{W}$, with $B_0 \supseteq \Gamma$. BANK growth is monotone:

$$B_t \subseteq B_{t+1}.$$

A conclusion enters BANK only when its certificate fragment is accepted by $\mathcal{V}$.

Definition 1.1 (Positive completeness for a fixed target). *A search procedure is positively complete for $(\mathfrak{F}, \Gamma, T)$ if*

$$\Gamma \vdash_{\mathfrak{F}} T \implies \exists t < \infty \text{ such that } T \in B_t.$$

Positive completeness does not assert finite termination when T is not derivable.

The distinction is fundamental. MAPQUEST can be designed to guarantee that an existing finite proof is eventually found without thereby becoming a total decider for arbitrary theoremhood.

2 Explicit WFF addresses from $\mathbb{N}$ onto $\mathbb{Q}$

Let $g : \mathcal{W} \rightarrow \mathbb{N}_0$ be an injective effective WFF numbering. Give each formula A the rational identity coordinate

$$a(A) = \frac{g(A)}{g(A) + 1} \in \mathbb{Q} \cap [0, 1).$$

Thus distinct WFF lines have distinct rational coordinates.

Suppose an explicit effective surjection

$$u : \mathbb{N}_+ \twoheadrightarrow \mathbb{Q} \cap [0, 1]$$

is available from a natural-to-rational enumeration. Define the least natural preimage of the formula address by

$$n(A) = \min\{n \in \mathbb{N}_+ : u(n) = a(A)\}.$$

Then every WFF has a finite natural-number address:

$$\boxed{A \longmapsto n(A) \in \mathbb{N}_+}.$$

Caution 2.1 (Identity is not proof distance). Neither

$$|a(A) - a(T)| \quad \text{nor} \quad |n(A) - n(T)|$$

is assumed to measure logical or proof-theoretic distance. The address identifies a formula. Target-directed proof distance is a separate object developed later.

For ordered tuples, use an effective bijective or injective pairing function. One convenient Cantor-style map on positive integers is

$$\pi(a, b) = \frac{(a + b - 1)(a + b - 2)}{2} + a,$$

and recursively

$$\pi_k(n_1, \dots, n_k) = \pi(n_1, \pi_{k-1}(n_2, \dots, n_k)).$$

Hence

$$\boxed{N(A_1, \dots, A_k) = \pi_k(n(A_1), \dots, n(A_k))}$$

is a finite address for every ordered finite premise tuple.

3 From premise addresses to complete inference addresses

Let $r \in \mathcal{R}$ be a rule of finite arity $k = \text{ar}(r)$. The intrinsic premise-address component is

$$X_r = \{r\} \times (\mathbb{Q} \cap [0, 1])^k,$$

and heterogeneous rule components are separated by the tagged coproduct

$$\boxed{X_{\mathcal{F}} = \bigsqcup_{r \in \mathcal{R}} \left(\{r\} \times (\mathbb{Q} \cap [0, 1])^{\text{ar}(r)} \right)}.$$

A WFF remains zero-dimensional; dimension belongs to the ordered premise slots of a rule application.

Production proof systems may require more than premise selection. Let

$$\Xi_r(A_1, \dots, A_k)$$

denote the effectively enumerable finite-description witnesses needed to instantiate r : substitutions, unifiers, metavariable assignments, side-condition data, or other finite rule-specific choices. A complete candidate inference is

$$c = (r; A_1, \dots, A_k; \xi), \quad \xi \in \Xi_r(A_1, \dots, A_k).$$

Give every rule an effective tag $\rho(r) \in \mathbb{N}_+$ and every finite witness an effective code $w(\xi) \in \mathbb{N}_+$. Then, for example,

$$\boxed{\text{Addr}(c) = \pi(\rho(r), \pi(N(A_1, \dots, A_k), w(\xi))) \in \mathbb{N}_+}.$$

Addressability principle. Every finite candidate inference in the declared effective proof system has a finite natural-number address. Consequently every finite proof certificate, being a finite list or DAG of such candidates, also has a finite natural-number encoding.

If

$$\Pi = (c_1, \dots, c_m)$$

is a finite proof, then one may encode it by repeated pairing:

$$\text{Addr}(\Pi) = \pi_m(\text{Addr}(c_1), \dots, \text{Addr}(c_m)).$$

The point is not that this code is computationally small. The point is that it is finite and effectively reachable by a fair enumeration.

4 Finite Hilbert boxes as rule-specific road atlases

For a k -ary rule, choose a resolution $p \geq 0$ and let

$$Q_{k,p} = \{0, \dots, 2^p - 1\}^k.$$

A discrete Hilbert traversal at that resolution is a bijection

$$H_{k,p} : \{0, \dots, 2^{pk} - 1\} \xrightarrow{\sim} Q_{k,p}.$$

Thus one complete pass visits every cell in the finite k -dimensional address box exactly once.

Every finite tuple $(n_1, \dots, n_k)$ lies in some finite box because for sufficiently large p ,

$$\max_i n_i < 2^p.$$

Therefore

$$\bigcup_{p=0}^{\infty} Q_{k,p} = \mathbb{N}_0^k.$$

Nested Hilbert boxes can therefore cover every finite premise tuple.

5 Why one Hilbert pass is not enough

A candidate can be visited before it becomes executable. Suppose MAPQUEST visits the cell for

$$A, B \xrightarrow{r} C$$

at time t , but only $A \in B_t$. The candidate cannot yet be verified because $B \notin B_t$. If the scheduler never revisits that cell after B is later proved, an existing proof can be missed despite complete coverage of the original pass.

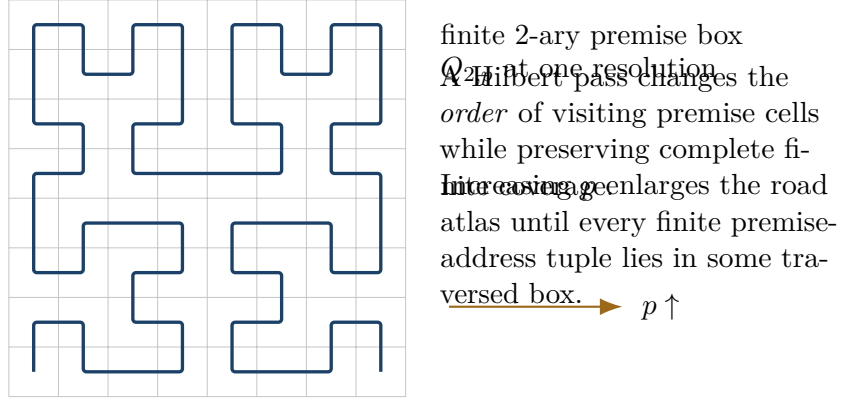


Figure 1: A schematic finite Hilbert road atlas for a binary rule. The drawn curve is illustrative; completeness depends on using a bijective finite Hilbert traversal at each chosen resolution.

Therefore completeness requires recurrence across two axes:

$$\boxed{\text{address resolution} \times \text{BANK epoch.}}$$

Let e index BANK epochs and p Hilbert resolutions. A diagonal dovetail such as

$$(0, 0), (0, 1), (1, 0), (0, 2), (1, 1), (2, 0), \dots$$

ensures that every finite resolution receives service after arbitrarily late BANK growth.

Definition 5.1 (Hilbert-fair recurrence). *A MAPQUEST scheduler is Hilbert-fair if every finite candidate inference c is reconsidered after any sufficiently late finite BANK stage at which all of its premises have become available. A stronger sufficient condition is*

$$\boxed{\forall c \quad \#\{t : S_t = c\} = \infty.}$$

The infinite-recurrence condition is stronger than necessary, but it makes the proof of positive completeness transparent.

6 Positive completeness theorem

We now state the central result in a form suitable for the MAPQUEST framework.

Theorem 6.1 (Hilbert-fair positive completeness). *Let $\mathfrak{F}$ be an effective finitary formal system satisfying:*

- (H1) *WFFs are effectively enumerable and each finite WFF has a finite effective address.*
- (H2) *Inference rules have finite arity and are finite or effectively enumerable.*
- (H3) *Every finite rule-instantiation witness space is effectively enumerable.*
- (H4) *Every finite candidate inference has a finite effective code and can be decoded effectively.*
- (H5) *The verifier V halts on every submitted finite candidate and is sound for the declared proof system.*

(H6) *Only verifier-accepted conclusions enter the monotone BANK.*

(H7) *The scheduler is Hilbert-fair across rule components, finite resolutions, witness codes, and BANK growth.*

Then, for every target T ,

$$\boxed{\Gamma \vdash_{\mathfrak{F}} T \implies \exists t < \infty : T \in B_t.}$$

Proof. Assume $\Gamma \vdash_{\mathfrak{F}} T$. By the ordinary definition of derivability in a finitary formal proof system, there is a finite proof DAG Π whose roots are axioms or hypotheses from Γ and whose final node is T . Topologically order the noninitial proof nodes as

$$C_1, \dots, C_m = T.$$

Let c_i be the finite inference application producing C_i from earlier nodes. By (H1)–(H4), each c_i has a finite address and lies in some finite Hilbert resolution with some finite witness code.

For c_1 , all premises are already in $B_0 \supseteq \Gamma$. Hilbert fairness guarantees a finite later reconsideration of c_1 . The verifier accepts the valid proof step, so C_1 enters BANK.

Assume inductively that every proof node preceding C_i has entered BANK after finite time. Because the proof order is topological, every premise of c_i is then present in BANK at some finite stage. Hilbert fairness guarantees a later finite reconsideration of c_i . Since c_i is a valid proof step and $\mathbf{V}$ is total and sound, C_i is accepted and deposited.

Finite induction through $i = 1, \dots, m$ yields $T = C_m \in B_t$ for some finite t . $\square$

Corollary 6.2 (Acceleration modules cannot destroy positive completeness if fairness is protected). *Suppose the hypotheses of the theorem hold and heuristic modules alter only candidate priority, region choice, or resource allocation while a nonstarvable Hilbert-fair channel remains active. Then Professor guidance, Dreamer, oracle advice, Tenneson1, and bounded Depth Charge may change discovery time but cannot remove eventual access to any existing finite proof.*

7 A crude finite bound for a particular proof

Let

$$\Pi = (c_1, \dots, c_m)$$

be a fixed valid proof and define

$$Z(\Pi) = \max_i \text{Addr}(c_i).$$

For a very simple scheduler that, at round s , reconsiders every address $z \leq s$, all proof-step addresses are permanently inside the horizon once $s \geq Z(\Pi)$. If proof depth is $d(\Pi)$ and one BANK-propagation layer can be enabled per full round, then a coarse bound has the form

$$\boxed{\tau_{\text{MQ}}(\Pi) \leq Z(\Pi) + d(\Pi) + O(1).}$$

Since $d(\Pi) \leq m$, an even coarser bound is

$$\tau_{\text{MQ}}(\Pi) \leq Z(\Pi) + m + O(1).$$

This is not claimed as a practical speed bound. The address magnitude can be enormous. Its purpose is to make the finiteness argument concrete.

8 The backward target cone: how MAPQUEST knows it is near T

Positive completeness explains why an existing proof cannot be lost. It does not yet explain how MAPQUEST should move quickly toward the fixed target. The missing structure is a backward target cone.

For a goal G , define

$$\text{Pred}(G) = \{(r, \sigma; P_1, \dots, P_k) : \text{Concl}(r, \sigma, P_1, \dots, P_k) = G\},$$

where σ stands for an admissible finite instantiation witness and any side conditions required by the rule. Each element of $\text{Pred}(G)$ is a candidate backward frame

$$P_1, \dots, P_k \implies G.$$

The frame is not a proof. It merely records a legal syntactic way in which G could be concluded if its premises were obtained and the verifier accepted the resulting step.

Starting with T , recursively expanding selected predecessor frames produces a budgeted backward dependency cone. The forward verified BANK grows upward from Γ . Search succeeds when verified BANK content repeatedly meets obligations in the backward cone until the final target frame closes.

9 Residual obligations as an exact near-target signal

For a backward frame

$$q = (r, \sigma; P_1, \dots, P_k \Rightarrow G),$$

define its unresolved-premise set

$$U_t(q) = \{P_i : P_i \notin B_t\}$$

and its missing-premise count

$$m_t(q) = |U_t(q)|.$$

Then

$$\boxed{m_t(q) = 0}$$

means the frame is ready for immediate verification, while

$$\boxed{m_t(q) = 1}$$

means the machine is one missing premise away from being able to attempt that route.

For the final target, define the direct residual count

$$\boxed{m_T(B_t) = \min_{q \in \text{Pred}(T)} m_t(q).}$$

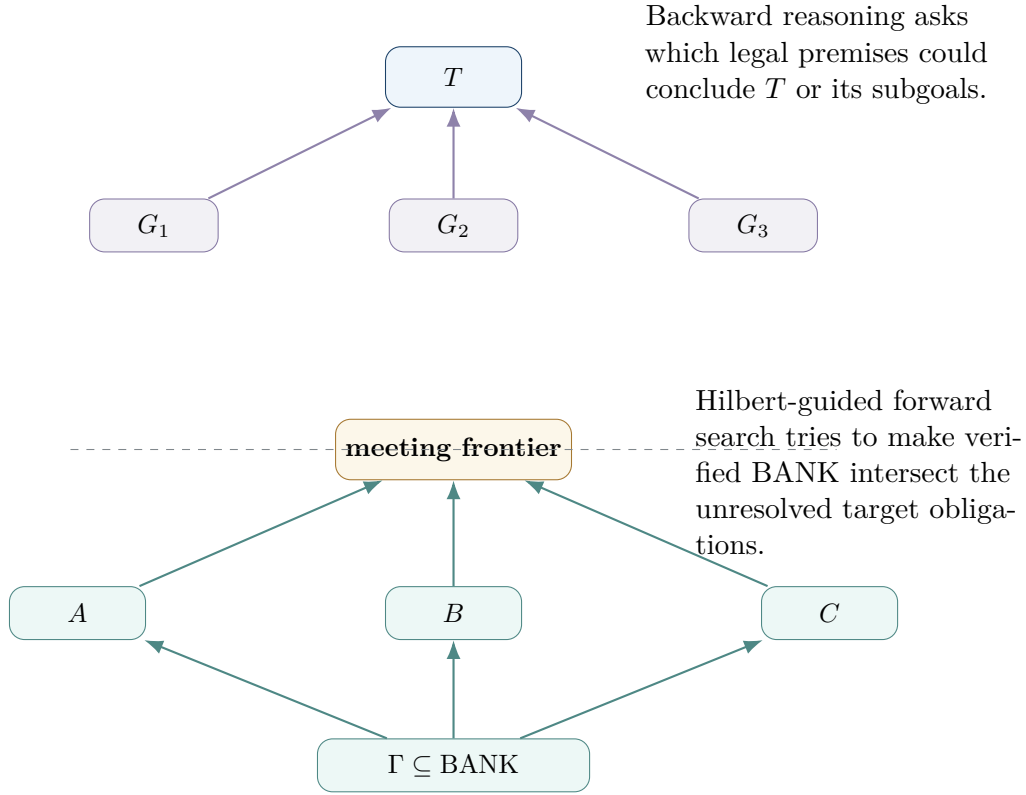


Figure 2: Bidirectional MAPQUEST geometry. The upper cone is a set of unverified backward obligations; the lower region is verifier-certified knowledge. The goal is not numerical address proximity but closure of the remaining legal proof obligations for T .

This gives a rigorous local notion of nearness. In particular,

$$m_T(B_t) = 0$$

means that at least one candidate final inference has all its premises already verified.

A richer recursive residual proof estimate is

$$\hat{h}_t(G) = 0 \quad \text{if } G \in B_t,$$

and otherwise

$$\hat{h}_t(G) = \min_{q=(P_1, \dots, P_k \Rightarrow G) \in \text{Pred}(G)} \left[c(q) + \Phi(\hat{h}_t(P_1), \dots, \hat{h}_t(P_k)) \right].$$

For depth-oriented guidance one may take

$$\Phi_{\text{depth}}(x_1, \dots, x_k) = \max_i x_i,$$

while a tree-cost heuristic may use

$$\Phi_{\text{tree}}(x_1, \dots, x_k) = \sum_i x_i.$$

The latter is only a heuristic for DAG proofs when branches can share lemmas.

The target potential is

$$\Psi_t = \widehat{h}_t(T).$$

Target progress is then observed when

$$\Psi_{t+1} < \Psi_t.$$

10 Waiting frames: the near-target acceleration mechanism

Suppose MAPQUEST has already discovered a candidate frame

$$A_1, \dots, A_k \implies G$$

but some premises are missing. Instead of forgetting the frame and waiting for a future global Hilbert sweep, store it in a waiting index keyed by its missing premises.

For a missing formula M , define

$$W(M) = \{q : M \in U_t(q)\}.$$

When M is later deposited into BANK, immediately update every frame in $W(M)$. If any such frame reaches

$$m_t(q) = 0,$$

move it to an immediate verification queue.

This gives a concrete mechanism for the situation in which the probe is close to T . Suppose

$$A, B, C \implies T, \quad A, C \in B_t, \quad B \notin B_t.$$

The frame waits on B . When B is proved, MAPQUEST need not rediscover the final tuple geometrically. It wakes the stored frame immediately, submits the final inference to the verifier, and, if accepted, deposits T .

Near-target principle. MAPQUEST is close to T not because a current formula has a rational address close to $a(T)$, but because a known legal target frame has few or low-cost unresolved premises. The decisive event is the closure of a verifier-checkable backward obligation.

11 The two-queue architecture

The cleanest way to combine guaranteed reachability with aggressive target guidance is to maintain two search queues.

Fair queue Q_F . This queue implements the canonical Hilbert-fair dovetail across rule tags, finite address resolutions, witness codes, and BANK epochs. It is the completeness backbone.

Guided queue Q_G . This queue contains candidates promoted by the target cone, residual proof estimates, Professor partial credit, Dreamer proposals, oracle guidance, Tenneson1, structural target features, and Depth Charge experiments.

The master scheduler must prevent starvation of Q_F . For example, a fixed service pattern

$$G, G, G, G, F, G, G, G, G, F, \dots$$

reserves 20% of raw expansion slots for the fair channel. More generally, any policy is acceptable if every finite fair-search obligation eventually receives service.

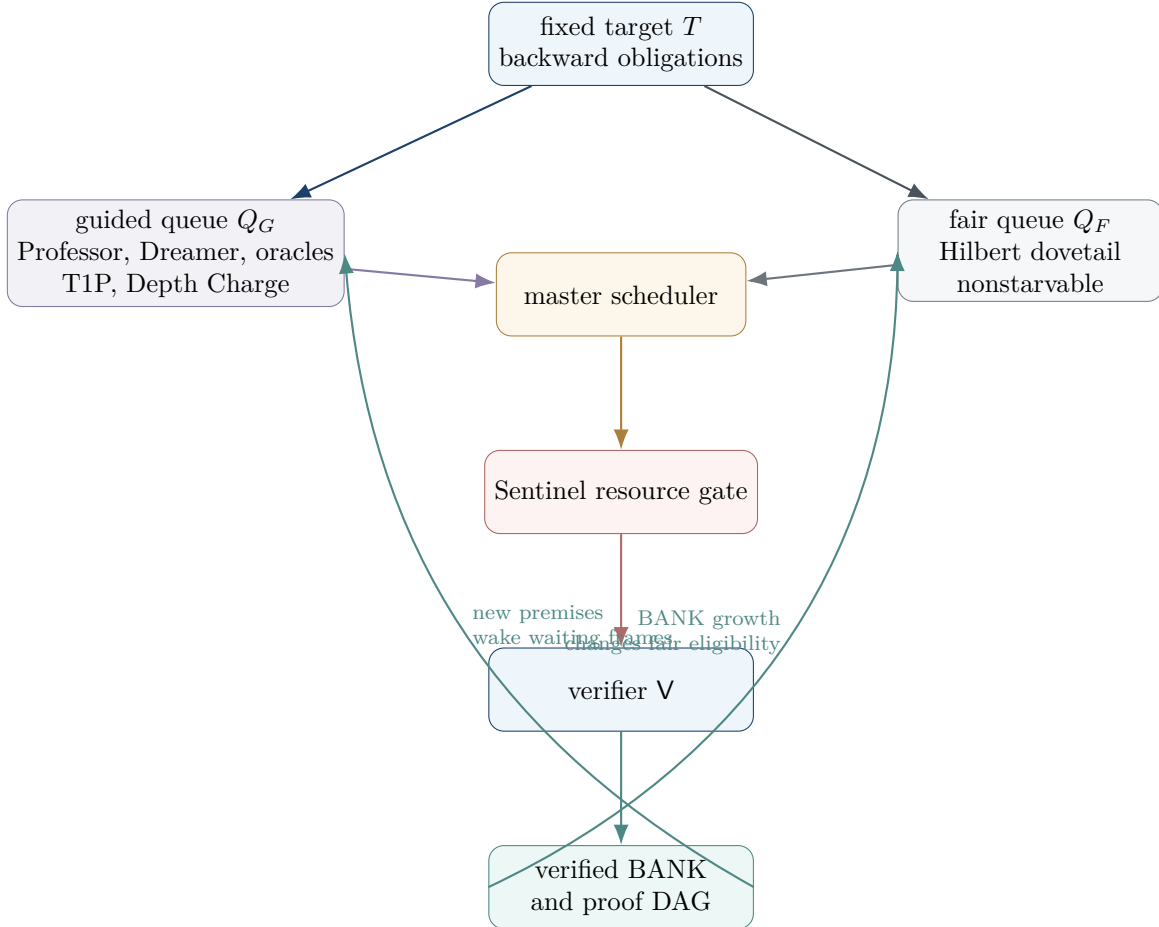


Figure 3: MAPQUEST as a protected two-queue system. The guided queue is allowed to be aggressive or wrong; the fair queue preserves access to every finite proof route. Both remain outside the trusted boundary until the verifier accepts a candidate.

12 Professor, Dreamer, and the oracle interfaces

These modules are most useful as target-directed accelerators, particularly when the backward cone has become narrow but one or more residual obligations remain difficult.

12.1 Professor as a target-progress estimator

For a candidate inference c with hypothetical verified successor state x^c , define the estimated target gain

$$\Delta_T(c) = \hat{h}_t(T) - \hat{h}_t^c(T).$$

The Professor may assign partial credit

$$P_t(c) \in [0, 1]$$

that is intended to correlate with positive $\Delta_T(c)$. A throttled contribution can be written

$$P_t^{\text{eff}}(c) = \lambda_t P_t(c), \quad 0 \leq \lambda_t \leq \lambda_{\max} < 1.$$

The Professor changes priority, not theoremhood.

12.2 Dreamer as route inventor

Dreamer is especially useful when the currently preferred backward decomposition of T contains a hard subgoal. It may propose an alternative decomposition, bridge lemma, representation, or rule sequence. Formally, one may treat it as

$$\mathcal{D}_t : \mathcal{H}_t \times B_D \rightarrow \mathcal{Q}_t,$$

where $\mathcal{H}_t$ is search history, B_D a bounded dream budget, and $\mathcal{Q}_t$ a finite set of untrusted proposals. Dreamer does not deposit conclusions into BANK.

12.3 Four oracle interfaces

A natural division of oracle roles near T is:

- O_1 : settlement-role guidance,
- O_2 : resource allocation among residual obligations,
- O_3 : strategy or Hilbert-region guidance,
- O_4 : candidate certificate proposal.

Even O_4 has no proof authority. If it proposes Π , acceptance still requires

$$V(\Gamma, T, \Pi) = 1.$$

13 A target-conditioned MAPQUEST score

For a legal candidate c , a practical guided score may combine estimated residual proof cost, resource cost, and advisory signals:

$$J_t(c; T) = c(c) + \hat{h}_t^c(T) - \lambda_P P_t(c) - \lambda_O O_t(c) + \lambda_R R_t(c),$$

where $O_t(c)$ is an allowed oracle-derived advisory score and $R_t(c)$ may penalize resource risk or redundant exploration. The guided queue approximately chooses

$$c_t \in \arg \min_{c \in Q_G} J_t(c; T).$$

This expression is an engineering navigation law, not part of the soundness proof. Positive completeness survives even if J_t is poor, provided the fair queue is protected.

14 Tenneson1 and Depth Charge in the completeness-preserving architecture

Tenneson1 can be interpreted as the guided controller that combines Hilbert-local traversal, target-conditioned scoring, creativity, and settlement-horizon estimates. Its job is to reduce search cost relative to the canonical fair enumerator.

Depth Charge is a bounded escape mechanism triggered by stagnation in target progress. One possible trigger is

$$\hat{h}_{t-w}(T) - \hat{h}_t(T) \leq \eta$$

for a declared window w and tolerance η . A Depth Charge may change rule components, Hilbert orientation, resolution, forward/backward emphasis, agent allocation, or creativity profile for a fixed budget B_{DC} . It may not bypass verification or consume the service share needed to keep Q_F nonstarvable.

Thus the architecture separates two claims:

$$\begin{array}{ll} Q_F & \Rightarrow \text{eventual reachability of every existing finite proof,} \\ Q_G & \Rightarrow \text{attempted acceleration toward this particular } T. \end{array}$$

15 Algorithm

The following pseudocode states the central mechanism independently of any specific implementation language.

Listing 1: Hilbert-fair MAPQUEST with target-cone guidance

```

1  INPUT: frozen environment F, hypotheses Gamma, target T, verifier V
2  STATE: verified BANK B := Gamma
3         proof DAG D
4         fair queue Q_F
5         guided queue Q_G
6         backward frames C_minus(T)
7         waiting index W
8
9  initialize explicit WFF/rule/witness addressing
10 initialize dovetailed Hilbert-fair enumerator Q_F
11 initialize target cone from T and seed Q_G
12
13 for t = 0,1,2,...:
14     refresh enabled waiting frames after BANK growth

```

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15
16   if T in B:
17       return verified dependency certificate for Gamma |- T
18
19   if scheduler owes a fair step:
20       c := next Hilbert-fair candidate from Q_F
21   else:
22       c := best target-conditioned candidate from Q_G
23
24   if Sentinel denies execution of c:
25       log and continue
26
27   decode c = (rule, premises, witness)
28
29   if some premise is missing from B:
30       register c in W(missing premise(s))
31       continue
32
33   construct candidate conclusion C and certificate fragment pi
34
35   if V(F, premises, C, pi) = ACCEPT:
36       if C is novel:
37           deposit C in B
38           add verified edge(s) to D
39           wake all frames waiting on C
40           update backward residuals for T
41           update Professor/Dreamer/oracle guidance
42   else:
43       log rejection
44
45   if target progress stagnates and Depth Charge is permitted:
46       run bounded Depth Charge; verify all retained mathematical outputs
47
48   return UNRESOLVED only when an external budget stops the run

```

16 Search trajectory versus proof certificate

The search trajectory can contain millions of failed or irrelevant addresses. The final proof certificate contains only verifier-accepted dependencies that lead to T . These are different mathematical objects.

If the final proof contains steps

$$P_{i1}, \dots, P_{ik_i} \xrightarrow{r_i, \xi_i} C_i,$$

then a Hilbert-addressed certificate can record

$$\Pi_T = [(r_i; \text{Addr}(P_{i1}), \dots, \text{Addr}(P_{ik_i}); \text{Addr}(\xi_i); \text{Addr}(C_i))]_{i=1}^m.$$

The final conclusion satisfies $C_m = T$. The verifier checks legal proof dependencies; it need not reproduce every failed Hilbert probe used during discovery.

17 What the theorem does not say

The positive-completeness result is intentionally precise and limited.

- It does not say MAPQUEST halts when T is not derivable.

- It does not say numerical address distance approximates proof distance.
- It does not say Hilbert order is faster than every other enumeration.
- It does not say Professor, Dreamer, or oracles are correct.
- It does not say an independence claim is certified merely because proof and refutation searches have run for a long time.
- It does not supply a useful polynomial or elementary complexity bound for arbitrary theorem search.

What it does say is important:

If a finite proof exists, a properly fair address-complete MAPQUEST cannot permanently miss it.

18 Why the natural-to-rational construction matters

Abstract computability theory can often establish countability without giving a useful explicit geometry. The natural-to-rational addressing layer does more operational work. It lets MAPQUEST attach concrete rational coordinates and canonical finite natural representatives to formulas, ordered premise tuples, and rule-specific action components. Those explicit coordinates can be indexed by discrete Hilbert traversals and stored in reproducibility logs.

The logical guarantee, however, requires the full chain

$$\begin{aligned}
 \mathbb{N} \twoheadrightarrow \mathbb{Q} &\implies \text{explicit finite addresses} \\
 &\implies \text{effective candidate decoding} \\
 &\implies \text{fair recurrence} \\
 &\implies \text{positive completeness.}
 \end{aligned}$$

The surjection supplies addressability. The fairness theorem supplies inevitability of discovery once a finite proof exists.

19 A sharpened research program

The theory now separates a theorem-level backbone from falsifiable efficiency hypotheses.

Backbone theorem. Under the stated effectivity, verifier, and fairness hypotheses,

$$\Gamma \vdash T \Rightarrow \text{finite-time discovery of a verified proof.}$$

Efficiency question. How much faster can target-conditioned guided search make discovery than the fair baseline on frozen held-out targets?

Near-target question. Do residual backward-obligation measures such as $m_T(B_t)$ and $\hat{h}_t(T)$ predict actual remaining proof cost?

Professor question. Is Professor partial credit positively correlated with true target gain Δ_T on held-out proof states?

Dreamer question. Does bounded counterfactual route generation produce alternative target decompositions that reduce verified discovery cost after accounting for Dreamer overhead?

Hilbert-locality question. Does the chosen formula/premise representation create real proof-relevance locality, or does Hilbert traversal behave like an arbitrary permutation? Random address permutation is a natural falsification control.

20 Conclusion

MAPQUEST becomes mathematically clearer when its two jobs are separated. The first job is nonnegotiable completeness infrastructure: every finite inference must have an effective finite address, every relevant address region must be traversable, candidates must be reconsidered after their premises become available, and only a total sound verifier may enlarge the verified BANK. Under those hypotheses, any existing finite proof of the fixed target T is reached after finite search time.

The second job is acceleration. Backward target cones identify what remains to be proved; waiting frames detect when a legal route is nearly complete; Hilbert traversals organize local rule-specific premise spaces; the Professor estimates target progress; Dreamer proposes alternate routes; oracle interfaces can guide settlement role, resources, strategy, or candidate certificates; Tenneson1 controls guided navigation; Depth Charge escapes stagnation; Sentinel controls execution resources. None of these modules changes what counts as a proof.

The resulting architecture can be summarized in one line:

address every road, revisit every road fairly,
prioritize roads that close T , and verify every step exactly.